

Solving the Coupled Pendulum SEP

We've successfully transformed the differential problem

$$\ddot{\mathbf{u}}(t) = -\mathbf{A} \dot{\mathbf{u}}(t)$$

into an algebraic problem involving matrix $\mathbf{A}$.
For our matrix

$$\mathbf{A} = \mathbf{M}^{-1} \mathbf{K} = \begin{bmatrix} \frac{1}{3} + \frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} + \frac{1}{3} \end{bmatrix}$$

let's use our knowledge of linear algebra to find our eigenvalue-eigenvector pairs.

We want to find scalars $\lambda = \omega^2$ and nonzero vectors $\vec{v} \in \mathbb{R}^2$ such that

$$\mathbf{A} \vec{v} = \lambda \vec{v}$$

$$\Leftrightarrow \mathbf{A} \vec{v} - \lambda \vec{v} = \vec{0}$$

$$\Leftrightarrow \mathbf{A} \vec{v} - \lambda \mathbf{I}_2 \vec{v} = \vec{0}$$

$$\Leftrightarrow (\mathbf{A} - \lambda \mathbf{I}_2) \vec{v} = \vec{0}$$

The only way we can find a nonzero $\vec{v} \in \mathbb{R}^2$ such that

$$(\mathbf{A} - \lambda \mathbf{I}_2) \vec{v} = \vec{0}$$

is if the matrix is singular

$\Rightarrow (\mathbf{A} - \lambda \mathbf{I}_2)$ has linearly dependent columns

$\Rightarrow \det(\mathbf{A} - \lambda \mathbf{I}_2) = 0 \leftarrow \text{characteristic equation}$

Consider the matrix

$$A - \lambda I_2 = \begin{bmatrix} \frac{g}{l} + \frac{k}{3m} & -\frac{k}{3m} \\ -\frac{k}{3m} & \frac{k}{3m} + \frac{g}{l} \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} \left(\frac{g}{l} + \frac{k}{3m}\right) - \lambda & -\frac{k}{3m} \\ -\frac{k}{3m} & \left(\frac{k}{3m} + \frac{g}{l}\right) - \lambda \end{bmatrix}_{2 \times 2}$$

Then, to find our eigenvalue-eigenvector pair $\lambda, \vec{v}$ is equivalent to the problem of finding λ such that

$$0 = \det(A - \lambda I_2) \quad \leftarrow \text{characteristic equation}$$

$$= \underbrace{\left[\left(\frac{g}{l} + \frac{k}{3m} \right) - \lambda \right]^2 - \left[-\frac{k}{3m} \right]^2}$$

characteristic polynomial: we want to find the roots

$$= \underbrace{\left[\left(\frac{g}{l} + \frac{k}{3m} \right) - \lambda \right]^2 - \left[\frac{k}{3m} \right]^2}$$

$$\Rightarrow \left[\left(\frac{g}{l} + \frac{k}{3m} \right) - \lambda \right]^2 = \left[\frac{k}{3m} \right]^2$$

$$\Rightarrow \sqrt{\left[\left(\frac{g}{l} + \frac{k}{3m} \right) - \lambda \right]^2} = \sqrt{\left[\frac{k}{3m} \right]^2}$$

both k (spring const)
and m (mass)
are positive

$$\Rightarrow \left| \left(\frac{g}{l} + \frac{k}{3m} \right) - \lambda \right| = \frac{k}{3m}$$

$$\Rightarrow \left| \lambda - \left(\frac{g}{l} + \frac{k}{3m} \right) \right| = \frac{k}{3m}$$

$$\Rightarrow \lambda - \left(\frac{g}{l} + \frac{k}{3m} \right) = \pm \frac{k}{3m}$$

There are 2 solutions to our characteristic equation (two distinct eigenvalues)

Eigenvalue 1: First Natural Frequency:

$$\lambda_1 = \left(\frac{g}{l} + \frac{k}{3m} \right) - \frac{k}{3m} = \frac{g}{l}$$

$$\Rightarrow \lambda_1 = \frac{g}{l} = \omega_1^2$$

$$\Rightarrow \omega_1 = \sqrt{\frac{g}{l}}$$

Eigenvalue 2: Second Natural Frequency:

$$\lambda_2 = \left(\frac{g}{l} + \frac{k}{3m} \right) + \frac{k}{3m}$$

$$= \frac{g}{l} + \frac{2k}{3m}$$

$$\Rightarrow \lambda_2 = \frac{g}{l} + \frac{2k}{3m} = \omega_2^2$$

$$\omega_2 = \sqrt{\frac{g}{l} + \frac{2k}{3m}}$$

$$\Rightarrow \lambda_1 = \frac{g}{l} \quad \text{and} \quad \lambda_2 = \frac{g}{l} + \frac{2k}{3m}$$

Eigenvector 1:

For $\lambda_1 = \frac{9}{2}$, we want to find nonzero $\vec{v}_1 \in \mathbb{R}^2$ such that

$$A \vec{v}_1 = \lambda_1 \vec{v}_1$$

$$\Leftrightarrow (A - \lambda_1 I_2) \vec{v}_1 = \vec{0}$$

$$\Leftrightarrow \begin{bmatrix} \left(\frac{19}{2} + \frac{1}{3}\right) \frac{9}{2} - \frac{1}{3} & -\frac{9}{2} - \frac{1}{3} \\ -\frac{1}{3} & \left(\frac{19}{3} + \frac{10}{2}\right) \frac{9}{2} \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{21} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Leftrightarrow \vec{v}_1 = \begin{bmatrix} v_{11} \\ v_{21} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Leftrightarrow A \vec{v}_1 = \lambda_1 \vec{v}_1, \text{ where}$$

$$\lambda_1 = \frac{9}{2} \text{ and } \vec{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\text{with } \omega_1^2 = \lambda_1 \Leftrightarrow \omega_1 = \sqrt{\frac{9}{2}}$$

Eigenvector 2:

For $\lambda_2 = \frac{g}{l} + \frac{2k}{m}$, we want to find nonzero $\vec{v}_2 \in \mathbb{R}^2$ such that

$$A \vec{v}_2 = \lambda_2 \vec{v}_2$$

$$\Leftrightarrow (A - \lambda_2 I_2) \vec{v}_2 = \vec{0}$$

$$\Leftrightarrow \begin{bmatrix} \left(\frac{g}{l} + \frac{2k}{m}\right) - \left(\frac{g}{l} + \frac{2k}{m}\right) & -\frac{1}{3}k \\ -\frac{1}{3}k & \left(\frac{2k}{m} + \frac{g}{l}\right) - \left(\frac{g}{l} + \frac{2k}{m}\right) \end{bmatrix} \begin{bmatrix} v_{12} \\ v_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} -\frac{1}{3}k & -\frac{1}{3}k \\ -\frac{1}{3}k & -\frac{1}{3}k \end{bmatrix} \begin{bmatrix} v_{12} \\ v_{22} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Leftrightarrow \vec{v}_2 = \begin{bmatrix} v_{12} \\ v_{22} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Leftrightarrow A \vec{v}_2 = \lambda_2 \vec{v}_2 \text{ where}$$

$$\lambda_2 = \frac{g}{l} + \frac{2k}{m} \quad \text{and} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\text{with } \omega_2^2 = \lambda_2 \Leftrightarrow \omega_2 = \sqrt{\frac{g}{l} + \frac{2k}{m}}$$

These two eigenvalue-eigenvector pairs can be grouped together into matrix form:

$$[A \vec{v}_1 \mid A \vec{v}_2] = [\lambda_1 \vec{v}_1 \mid \lambda_2 \vec{v}_2]$$

$$\Leftrightarrow A [\vec{v}_1 \mid \vec{v}_2] = [\vec{v}_1 \mid \vec{v}_2] \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$$

$$\Leftrightarrow [A]_{2 \times 2} \cdot [V]_{2 \times 2} = [V]_{2 \times 2} \cdot [\Lambda]_{2 \times 2}$$



$$\begin{bmatrix} 1 & & \\ p/3 & + & \\ 3/2 & & \end{bmatrix}$$

$$\begin{bmatrix} 3/2 & & 1 \\ + & & 3/2 \\ p/3 & & \end{bmatrix}$$

$$\begin{bmatrix} 1 & & \\ - & & \\ - & & \end{bmatrix}$$

$$\begin{bmatrix} 1 & & \\ - & & \\ - & & \end{bmatrix}$$

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$$\begin{bmatrix} 1 & & \\ - & & \\ - & & \end{bmatrix}$$

$$\begin{bmatrix} 1 & & \\ - & & \\ - & & \end{bmatrix}$$

$$\begin{bmatrix} 1 & & \\ 0 & & \\ 0 & & \end{bmatrix}$$

$$\begin{bmatrix} p/3 & & 0 \\ + & & \\ 3/2 & & \end{bmatrix}$$